ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: semiconductors and transistor switching, Boolean algebra and combinational logic, flip-flops and counters, analogue-to-digital conversion and sensor interfacing (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Adding impurities to pure silicon to change its electrical properties is called …

A etching. B doping. C soldering. D plating.

1.1.2 When a transistor used as a switch is fully ON, it is said to be in …

A cut-off. B breakdown. C reverse bias. D saturation.

1.1.3 According to De Morgan's theorem, the complement of $(A \cdot B)$ is …

A $\bar{A} \cdot \bar{B}$. B $\bar{A} + \bar{B}$. C $A + B$. D $A \cdot \bar{B}$.

1.1.4 The Boolean expression $A + A \cdot B$ simplifies to …

A B. B $A \cdot B$. C A. D 1.

1.1.5 A circuit whose output depends only on its present inputs is a …

A combinational logic circuit. B sequential logic circuit.
C memory circuit. D counter.

1.1.6 The basic memory element that stores one bit is a …

A decoder. B flip-flop. C multiplexer. D resistor.

1.1.7 A 10-bit analogue-to-digital converter (ADC) can give … different output values.

A 1 000 B 512 C 10 D 1 024

1.1.8 A circuit that selects one of several inputs and sends it to a single output is a …

A multiplexer. B demultiplexer. C encoder. D comparator.

1.1.9 A 4-bit binary counter counts from 0 to …

A 16. B 4. C 15. D 8.

1.1.10 The two terminals of a diode are the …

A base and emitter. B gate and drain.
C anode and cathode. D source and collector.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 An N-type semiconductor has extra free electrons.

1.2.2 A forward-biased silicon diode has a voltage drop of about 0,7 V.

1.2.3 NAND and NOR gates are called universal gates because any logic function can be built from them.

1.2.4 In a D flip-flop, the output Q copies the D input on the active clock edge.

1.2.5 A MOSFET is controlled by base current.

1.2.6 A seven-segment display needs a decoder to show decimal digits from binary input.

1.2.7 A Karnaugh map is used to simplify Boolean expressions.

1.2.8 An analogue sensor gives only two values, 0 and 1.

1.2.9 Pull-up resistors keep a digital input at a known HIGH level when a switch is open.

1.2.10 A JK flip-flop toggles when J = 1, K = 1 and a clock pulse is applied.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Ratio of collector current to base current	A	current gain (h_{FE})
1.3.2	Diode that keeps a constant voltage when reverse-biased	B	sampling rate
1.3.3	Component that switches using light and isolates two circuits	C	resolution
1.3.4	Signal that controls when a flip-flop changes state	D	decoder
1.3.5	Smallest change in input voltage an ADC can detect	E	clock
1.3.6	Number of samples taken per second	F	Zener diode
1.3.7	Group of flip-flops that shifts data along one place per clock pulse	G	optocoupler
1.3.8	Unwanted multiple signals produced when a mechanical switch is pressed	H	shift register
1.3.9	Circuit that compares two voltages and gives a digital output	I	comparator
1.3.10	Converts a binary code into one active output line	J	contact bounce

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: SEMICONDUCTORS AND TRANSISTOR SWITCHING

2.1 Explain the difference between P-type and N-type semiconductor material. (4)

2.2 A microcontroller output (5 V) must switch a 12 V relay coil that needs 60 mA, using an NPN transistor with h_{FE} = 100. Assume V_{BE} = 0,7 V.

2.2.1 Calculate the base current needed. (2)

2.2.2 Calculate the maximum value of the base resistor. (3)

2.2.3 Explain why a slightly smaller standard resistor (e.g. 4,7 kΩ) is normally chosen. (2)

2.2.4 Draw the circuit, including the flyback diode. (5)

2.3 Give TWO advantages of using a MOSFET instead of a bipolar transistor to drive a robot's motor. (2)

2.4 Explain why an optocoupler is used between a microcontroller and a high-voltage motor circuit. (2)

[20]

QUESTION 3: BOOLEAN ALGEBRA AND COMBINATIONAL LOGIC

3.1 Use Boolean algebra to simplify: (a) $X = A \cdot B + A \cdot \bar{B}$ (b) $Y = \bar{A} \cdot \bar{B} + \bar{A} \cdot B + A \cdot B$ (6)

3.2 Draw the logic circuit for Y in its simplified form. (2)

3.3 A robot's warning buzzer (Z) must sound when at least TWO of its three sensors (A, B and C) detect an obstacle.

3.3.1 Draw up the truth table. (4)

3.3.2 Write the simplified Boolean expression for Z. (3)

3.3.3 Draw the logic circuit using AND and OR gates. (3)

3.4 Show how a NOT gate and an AND gate can each be built using only NAND gates. (2)

[20]

QUESTION 4: FLIP-FLOPS AND COUNTERS

4.1 Draw the truth table of an SR (set–reset) latch built with NOR gates and state which input combination is not allowed. (5)

4.2 Explain how a D flip-flop can be used to store the state of a push button. (3)

4.3 Explain the difference between an asynchronous (ripple) counter and a synchronous counter. (4)

4.4 How many flip-flops are needed to count from 0 to 9 (a decade counter)? Explain. (3)

[15]

QUESTION 5: ADC AND SENSOR INTERFACING

5.1 A robot's Arduino has a 10-bit ADC with a reference voltage of 5 V.

5.1.1 Calculate the resolution of the ADC in millivolts. (2)

5.1.2 A temperature sensor gives 1,25 V. Calculate the digital value read by the ADC. (2)

5.1.3 The ADC reads 818. Calculate the input voltage. (2)

5.2 An LDR and a 10 kΩ resistor form a voltage divider across 5 V. In bright light the LDR is 1 kΩ. Calculate the output voltage across the 10 kΩ resistor. (3)

5.3 Explain how contact bounce can be removed by hardware or by software. (3)

5.4 Explain the difference between an analogue and a digital sensor, giving ONE robot example of each. (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER